CAPSTONE PROJECT REPORT

AI LECTURE NOTES SUMMARISER AND MIND-MAP CREATOR

A project on using Artificial Intelligence tools to summarise learning materials, create mind maps, and verify AI-generated information.

Academic Project Report

Important note before submission

This is a ready-to-edit project report draft. The lecture summaries, sample prompts, mind-map outlines, takeaways and viva questions are prepared as project content. To honestly complete the experiment, you must upload the same source material to NotebookLM, Gemini and Claude, save screenshots of their actual outputs, and fill in the comparison scores in Section 6. Do not claim that peer feedback was collected until classmates have actually reviewed your mind map.

Use the official/open educational sources listed in the References section. Check the exact PDF or lecture you select, note its title and date, and keep a copy or link for your evidence.

Abstract

This project studies how Artificial Intelligence (AI) tools can convert long educational materials into short and useful learning resources. Three subjects are used: Introduction to Artificial Intelligence, Cloud Computing, and Fundamentals of Cyber Security. The same source material for each subject is processed using NotebookLM, Gemini and Claude. Their summaries are compared for correctness, completeness, clarity and coverage. Important concepts are arranged into hierarchical mind-map outlines and can be designed using Canva or Miro. Each subject also contains ten key takeaways and five viva questions with answers. Finally, the outputs are checked against the original learning material to identify missing, incorrect or outdated details. The project highlights that AI can support learning, but human verification is necessary.

1. Introduction

Students often receive lengthy lecture notes, PDFs and video lectures. Reading all the material again during revision can take considerable time. AI tools can help by identifying main ideas, explaining difficult terms, creating summaries and suggesting questions. A mind map presents the relationship between topics in a visual and organised form. This project combines AI summarisation, prompt engineering, visual content creation and fact-checking in one workflow.

2. Problem statement

Long lectures and reading materials may be difficult to revise quickly. The project asks: Can AI convert long lectures and reading material into clear summaries and mind maps without losing important meaning or adding incorrect information?

3. Objectives

• Select three publicly available educational lecture resources.

• Generate summaries of the same material using NotebookLM, Gemini and Claude.

• Compare the AI outputs for completeness, correctness, clarity and usefulness.

• Create a hierarchical outline and a visual mind map for each subject.

• Prepare ten key takeaways and five viva questions for each subject.

• Identify missing, inaccurate or outdated AI-generated information.

• Record prompts, verification steps and peer feedback.

• Explain privacy, copyright, transparency and the limitations of AI.

4. Tools required

Tool	Purpose	Website
NotebookLM	Summarise and ask questions from uploaded source documents	https://notebooklm.google.com/
Gemini	Generate and simplify summaries; create question lists	https://gemini.google.com/
Claude	Generate an independent summary and structured outline	https://claude.ai/
Canva or Miro	Design the final mind maps	https://www.canva.com/ | https://miro.com/
Google Docs or MS Word	Prepare and format the report	https://docs.google.com/

The project uses at least four tools. Free-tier limits and features may change, so check each service before use.

5. Source materials selected

Use these open educational resources as starting points. Select one introductory unit or lecture from each source and record the exact title, author/institution and access date in your final copy.

Lecture	Open educational source	Selected scope
1. Artificial Intelligence	MIT OpenCourseWare – Artificial Intelligence lecture notes: https://ocw.mit.edu/courses/6-034-artificial-intelligence-spring-2005/pages/lecture-notes/	Chapter 1: Introduction to AI
2. Cloud Computing	CUNY Open Educational Resources – Lecture Notes on Cloud Computing: https://academicworks.cuny.edu/qc_oers/129/	Introduction, cloud infrastructure, service concepts and resource model
3. Cyber Security	Purdue University – Computer and Network Security lecture notes: https://engineering.purdue.edu/kak/compsec/Lectures.html	Introductory security vocabulary and basic security concepts

Additional source option for AI: IGNOU eGyanKosh, Unit 1 – Introduction to Artificial Intelligence: http://egyankosh.ac.in/handle/123456789/86439

Additional source option for Cloud Computing: Cambridge Computer Science course materials: https://www.cl.cam.ac.uk/teaching/2526/CC/materials.html

6. Methodology / workflow

• Find one publicly accessible lecture or PDF for each of the three subjects.

• Record the source title, author/institution, URL and date accessed.

• Use the same source file or same selected lecture content in NotebookLM, Gemini and Claude.

• Give each AI tool the same prompt so that the comparison is fair.

• Save each output and screenshot; do not edit the original AI output before saving evidence.

• Compare the outputs using the evaluation table below.

• Create a hierarchical outline from the verified summary.

• Design one mind map per subject in Canva or Miro.

• Prepare ten key takeaways and five viva questions per subject.

• Check every important statement against the original source and record corrections.

• Ask classmates to review one mind map and record their real feedback.

• Prepare the final report and demonstrate one complete example.

6.1 Common prompt used for all three AI tools

PROMPT: “Read the attached lecture material carefully. Create a student-friendly summary using simple English. Use clear headings and bullet points. Include the main concepts, important definitions, examples and relationships between topics. Do not add facts that are not supported by the source. If information is missing or unclear, say so. At the end, list important points that a student should verify in the original material.”

6.2 Prompt for mind-map outline

PROMPT: “Using only the verified summary, create a hierarchical mind-map outline. Put the main topic at the centre, then create 4–6 main branches and smaller sub-branches. Use short keywords rather than long sentences. Keep the relationship between ideas clear. Do not introduce new facts.”

6.3 Prompt for key takeaways and viva

PROMPT: “Prepare exactly 10 important key takeaways and 5 viva questions with short answers from the supplied lecture material. Use simple student-friendly English. Keep answers accurate and do not add unsupported information.”

7. Lecture 1 – Introduction to Artificial Intelligence

7.1 Summary

Artificial Intelligence (AI) is a field of computer science that develops systems able to perform tasks that normally require human intelligence. Examples include recognising speech, understanding language, identifying patterns, making recommendations and solving problems. AI systems use data, algorithms and computing power. Machine learning is a part of AI in which a system learns patterns from examples. Deep learning is a type of machine learning that uses multi-layer neural networks. AI is used in education, healthcare, transport, banking and customer services. AI results must be checked because systems can make mistakes, reflect bias or produce unsupported information.

7.2 Mind-map outline

ARTIFICIAL INTELLIGENCE (central topic)

■■ Meaning: computer systems performing intelligent tasks

■■ Main areas: machine learning, deep learning, natural language processing, computer vision

■■ Working: data → algorithm/model → output → evaluation

■■ Applications: education, healthcare, transport, banking

■■ Challenges: bias, privacy, errors, explainability and responsible use

7.3 Ten key takeaways

• AI enables computers to perform tasks that usually require human intelligence.

• AI systems use data and algorithms to produce results.

• Machine learning is a branch of AI.

• Machine learning identifies patterns from examples or data.

• Deep learning uses multi-layer neural networks.

• Natural language processing helps computers work with human language.

• Computer vision helps systems interpret images and videos.

• AI is used in many fields, including education and healthcare.

• AI outputs can contain errors or bias.

• Human checking and responsible use are important.

7.4 Five viva questions with answers

• 1. What is Artificial Intelligence?

Answer: AI is a branch of computer science that enables machines to perform tasks that normally require human intelligence.

• 2. What is Machine Learning?

Answer: Machine Learning is a part of AI in which systems learn patterns from data.

• 3. What is Deep Learning?

Answer: Deep Learning is a type of machine learning that uses multi-layer neural networks.

• 4. Mention two applications of AI.

Answer: AI is used in healthcare and education.

• 5. Why should AI outputs be checked?

Answer: AI can produce incorrect, biased or unsupported information.

8. Lecture 2 – Cloud Computing

8.1 Summary

Cloud computing is the delivery of computing services such as servers, storage, databases, networking and software over the internet. Instead of buying and maintaining all computing hardware locally, an organisation can use resources provided by a cloud provider. Common service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS). Deployment models include public, private, hybrid and community clouds. Important characteristics include on-demand access, resource sharing, scalability and measured usage. Cloud computing can reduce the need to maintain physical infrastructure, but users must consider security, privacy, availability, cost and the shared responsibility between provider and customer.

8.2 Mind-map outline

CLOUD COMPUTING (central topic)

■■ Meaning: computing services delivered over the internet

■■ Service models: IaaS, PaaS, SaaS

■■ Deployment models: public, private, hybrid, community

■■ Characteristics: on-demand, shared resources, scalability, measured service

■■ Uses: storage, hosting, databases, software access and backup

■■ Concerns: security, privacy, cost, availability and provider dependency

8.3 Ten key takeaways

• Cloud computing provides computing resources through the internet.

• Users can access cloud services when needed.

• IaaS provides virtualised infrastructure resources.

• PaaS provides a platform for developing and deploying applications.

• SaaS provides ready-to-use software over the internet.

• Public cloud services are offered to multiple customers.

• Private cloud resources are dedicated to one organisation.

• Hybrid cloud combines private and public cloud environments.

• Scalability allows resources to increase or decrease based on demand.

• Security, privacy, cost and availability must be considered.

8.4 Five viva questions with answers

• 1. What is cloud computing?

Answer: It is the delivery of computing services such as storage, servers and software over the internet.

• 2. Expand IaaS.

Answer: Infrastructure as a Service.

• 3. What does PaaS provide?

Answer: It provides a platform to develop, test and deploy applications.

• 4. What is SaaS?

Answer: Software as a Service provides ready-to-use software through the internet.

• 5. What is a hybrid cloud?

Answer: It combines private cloud and public cloud environments.

9. Lecture 3 – Fundamentals of Cyber Security

9.1 Summary

Cyber security is the practice of protecting computers, networks, applications and data from unauthorised access, damage or disruption. A basic security model is the CIA triad: Confidentiality, Integrity and Availability. Confidentiality means information is accessible only to authorised people. Integrity means information remains accurate and is not changed without permission. Availability means systems and information are accessible when needed. Common risks include weak passwords, phishing, malware, unpatched software and unsafe network settings. Basic protection includes strong unique passwords, multi-factor authentication, software updates, backups, access control, encryption and user awareness. Security is an ongoing process that includes prevention, detection and response.

9.2 Mind-map outline

CYBER SECURITY (central topic)

■■ Goal: protect systems, networks and information

■■ CIA triad: confidentiality, integrity, availability

■■ Threats: phishing, malware, weak passwords, unpatched software

■■ Protection: MFA, updates, backups, encryption, access control

■■ Process: prevent → detect → respond → recover

■■ People: awareness, safe habits and responsible reporting

9.3 Ten key takeaways

• Cyber security protects devices, networks, applications and data.

• Confidentiality prevents unauthorised information access.

• Integrity protects information from unauthorised changes.

• Availability ensures systems and data can be accessed when required.

• The CIA triad is a basic information-security model.

• Phishing attempts to trick users into revealing information or taking unsafe actions.

• Weak passwords increase the risk of account compromise.

• Multi-factor authentication adds another layer of account protection.

• Updates and backups help improve security and recovery.

• Users should report suspicious activity through approved channels.

9.4 Five viva questions with answers

• 1. What is cyber security?

Answer: It is the practice of protecting systems, networks and data from digital threats.

• 2. What does CIA stand for?

Answer: Confidentiality, Integrity and Availability.

• 3. Define confidentiality.

Answer: It means information is available only to authorised users.

• 4. What is phishing?

Answer: It is a deceptive message or website designed to trick people into sharing information or taking an unsafe action.

• 5. Mention two cyber security practices.

Answer: Use multi-factor authentication and install software updates.

10. Comparison of AI tools

Run the same prompt and use the same source material in all three tools. The table below is an evaluation sheet; enter scores only after you inspect the actual outputs. Use a score from 1 (poor) to 5 (excellent).

Evaluation criteria	NotebookLM (1–5)	Gemini (1–5)	Claude (1–5)
Correctness			
Completeness			
Clarity / simple language			
Organisation / headings			
Usefulness for revision			
Total out of 25			

Suggested interpretation: the tool with the highest verified score is the most suitable for this particular source and task. Do not assume one tool is always best; results can change with the topic, prompt and source.

10.1 Evidence to attach

• Screenshot of the same source uploaded to each AI tool (where permitted).

• Screenshot or saved copy of each tool's summary.

• Screenshot of the prompt used.

• Completed scoring table with brief reasons for each score.

• One example where you corrected or supplemented an AI output.

11. Missed-points and correction analysis

Use this table after checking each AI summary against the original lecture. The examples below show the type of issue to look for; they are not claims that a particular tool made these mistakes.

Possible issue to check	Why it matters	Correction method
AI leaves out a definition	The summary may be incomplete	Add the definition from the original source
AI mixes up IaaS, PaaS and SaaS	Service models have different purposes	Compare each definition with the lecture
AI explains only two parts of the CIA triad	The security concept becomes incomplete	Check confidentiality, integrity and availability
AI adds an example not found in the source	It may be unsupported by the selected lecture	Label it as extra information or remove it
AI uses outdated details	Technology and services can change	Check current official documentation and date the correction

Record actual findings here:

Lecture / AI tool	Missed or incorrect point	Evidence in source	Correction made

12. Verification report

For each summary, check the following:

• Are the main headings and concepts present?

• Are definitions consistent with the original lecture?

• Are names, dates, numbers and technical terms correct?

• Are examples supported by the source?

• Has the AI clearly identified uncertainty or missing information?

• Is any information outdated or outside the lecture scope?

• Have corrections been recorded with a source reference?

Verification result template: Lecture ___ / AI tool ___ / Number of issues found ___ / Corrections made ___ / Checked by ___ / Date ___.

13. Before-and-after analysis

Before using AI	After using AI and verification
Long lecture or PDF that takes time to revise	Short structured summary with headings
Important ideas may be spread across many pages	Main ideas are grouped by topic
Relationships between concepts may be difficult to see	Mind map shows main branches and sub-branches
Students must prepare revision questions themselves	Draft viva questions are generated and checked
AI output may contain unsupported details	Output is compared with the source and corrected

AI does not replace reading or understanding the original material. It is a support tool for revision.

14. Peer feedback

Show one completed mind map to at least three classmates. Ask them to review clarity, correctness, layout and usefulness. Enter only the feedback you actually receive.

Reviewer (initials only)	What was clear?	What can be improved?	Overall suggestion

Summary of feedback: _____

15. Automation / workflow

The project follows a repeatable semi-automated workflow. The AI tools generate draft content, while the student checks and approves it.

• Input: lecture PDF or publicly available lecture material.

• Processing: upload the same source and enter a standard prompt in each AI tool.

• Output 1: summaries from NotebookLM, Gemini and Claude.

• Output 2: verified outline, mind map, takeaways and viva questions.

• Quality check: compare generated statements with the original source.

• Storage: save approved outputs, prompts, screenshots and correction notes in organised folders.

• Final output: project report and presentation.

Workflow diagram (text form):

Lecture PDF / video → AI summarisation (3 tools) → Compare outputs → Verify against source → Mind-map outline → Canva/Miro design → Takeaways + viva → Final report

16. Prompt library

Summary prompt

Summarise the attached lecture in simple English. Use headings and bullet points. Include key definitions and examples. Do not add unsupported facts. Mention unclear or missing information.

Comparison prompt

Compare these summaries with the original lecture. Identify missing concepts, incorrect statements, unclear explanations and unsupported additions. Give the source section that should be checked.

Mind-map prompt

Create a hierarchical mind-map outline from the verified summary. Use one central topic, 4–6 main branches and short sub-branches. Use keywords, not paragraphs.

Takeaway prompt

Create exactly 10 key takeaways from the verified lecture. Each point must be short, accurate and useful for revision.

Viva prompt

Create exactly 5 viva questions with short answers from the lecture. Include basic definitions and concept-understanding questions.

Simplification prompt

Rewrite this explanation in simple student-friendly English without changing its technical meaning.

17. Ethical considerations and limitations

• Privacy: Do not upload personal, confidential or restricted college information.

• Copyright: Use permitted educational resources, respect licences and acknowledge the original author or institution.

• Honest disclosure: State which AI tools were used and which parts were AI-assisted.

• Accuracy: AI can generate incorrect, incomplete or unsupported information; verify important claims.

• Bias: AI output may reflect bias in its training data or source material.

• Human responsibility: The student is responsible for checking, editing and understanding the final work.

• Tool limits: Free plans may have file-size, usage or feature limits; these may change over time.

18. Result and discussion

The project demonstrates a structured way to transform lecture material into revision resources. The three AI tools can be assessed using the same source and prompt. The mind maps organise concepts visually, while takeaways and viva questions support revision. The comparison table and correction log provide evidence of the strengths and limitations of AI-generated content. The final conclusion about which tool performed best must be based on the student's completed evaluation scores and saved outputs.

19. Conclusion

AI tools can help students summarise long educational materials, organise ideas and prepare revision questions. NotebookLM, Gemini and Claude can be compared by giving them the same lecture and prompt. Canva or Miro can be used to turn verified outlines into mind maps. However, AI-generated content should not be accepted without checking. Source verification, privacy, copyright and honest disclosure are essential. Therefore, AI is most useful as a learning assistant when combined with human judgement.

20. References

• MIT OpenCourseWare, Artificial Intelligence – Lecture Notes. https://ocw.mit.edu/courses/6-034-artificial-intelligence-spring-2005/pages/lecture-notes/

• Jun Li, CUNY Queens College, Lecture Notes on Cloud Computing (Winter 2026). https://academicworks.cuny.edu/qc_oers/129/

• Avi Kak, Purdue University, Computer and Network Security – Lecture Notes. https://engineering.purdue.edu/kak/compsec/Lectures.html

• Google NotebookLM. https://notebooklm.google.com/

• Google Gemini. https://gemini.google.com/

• Anthropic Claude. https://claude.ai/

• Canva. https://www.canva.com/

• Miro. https://miro.com/

Appendix A – Final submission checklist

• ■ Three source lecture materials and their details are recorded.

• ■ Three summaries are prepared.

• ■ Actual outputs from NotebookLM, Gemini and Claude are saved.

• ■ AI comparison scores and reasons are completed.

• ■ Three mind maps are created.

• ■ Thirty key takeaways and fifteen viva questions are included.

- ■ Missed-point analysis contains actual checked examples.
- ■ Peer feedback is collected and recorded.
- ■ Prompt library is included.
- ■ Verification report and corrections are completed.
- ■ Before-and-after analysis is included.
- ■ Final demonstration slides are prepared.